MEN 472

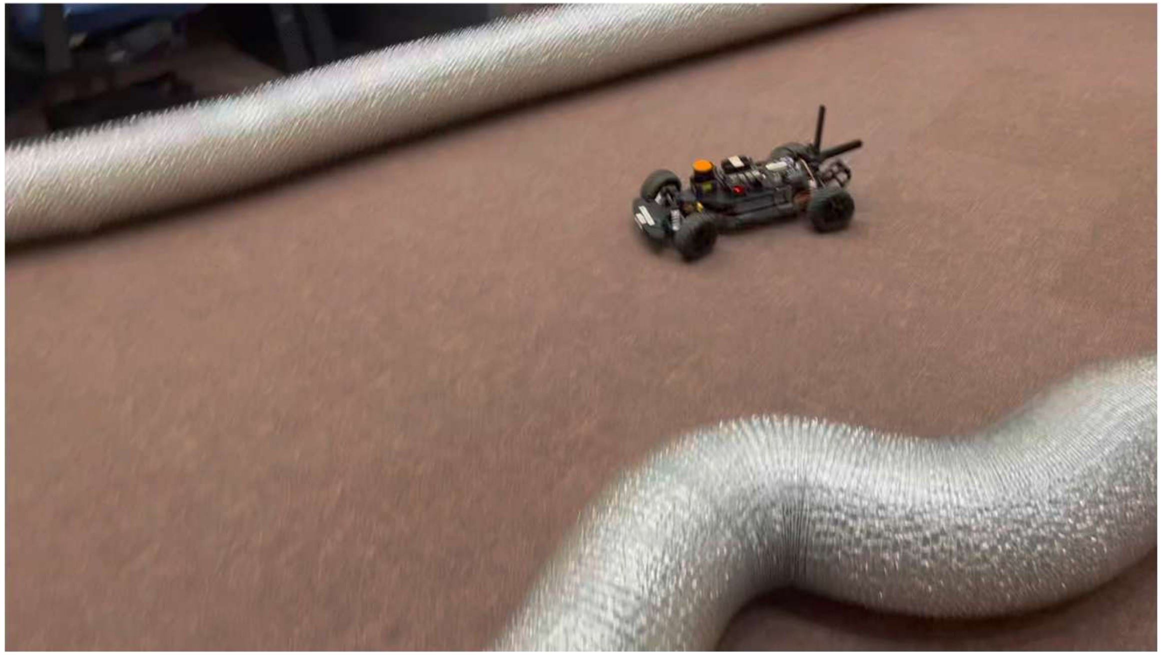
Creating Autonomous Vehicle

Week6-1
Gap-following Practice Session

2026.04.07

ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

Practice Goal

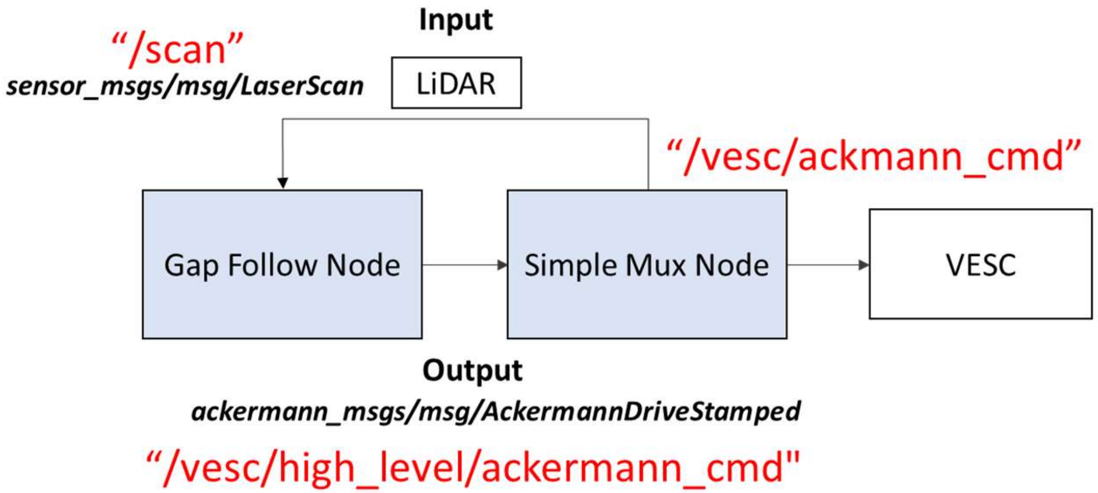


- The goal of this lab is to develop a reactive algorithm for race car to avoid obstacles.
- Find and follow the drivable gap.

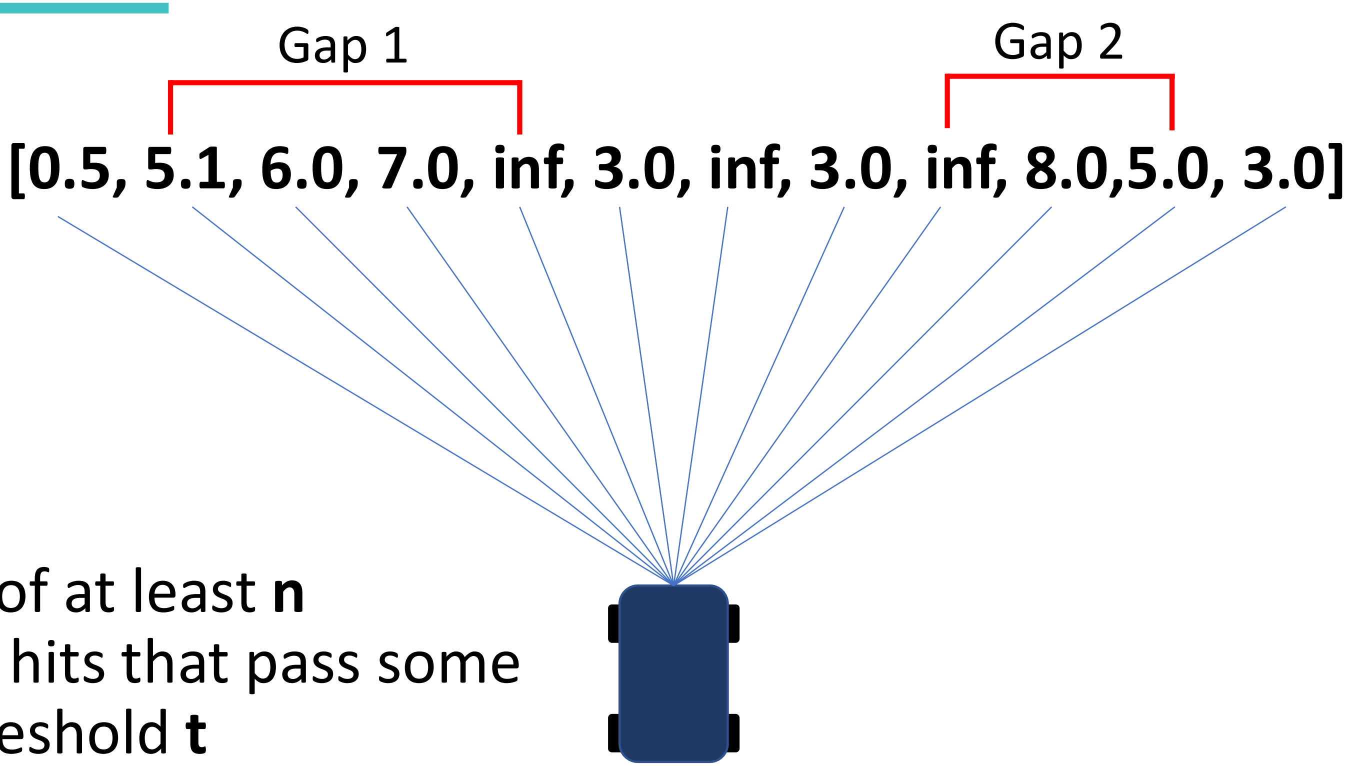
Overview

- For this lab, you will make a '**Gap Following Node**' that should make the car maintain a constant distant from the wall while moving forward
- You will need to first <u>subscribe to the scan topic</u> and <u>find the max length "gap"</u> with the ***LaserScan*** messages
- We are sending ***AckermannDriveStamped*** messages with <u>desired velocity and steering angle</u>

```python
import math
import numpy as np
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import LaserScan
from nav_msgs.msg import Odometry
from ackermann_msgs.msg import AckermannDriveStamped
```



Follow the Gap

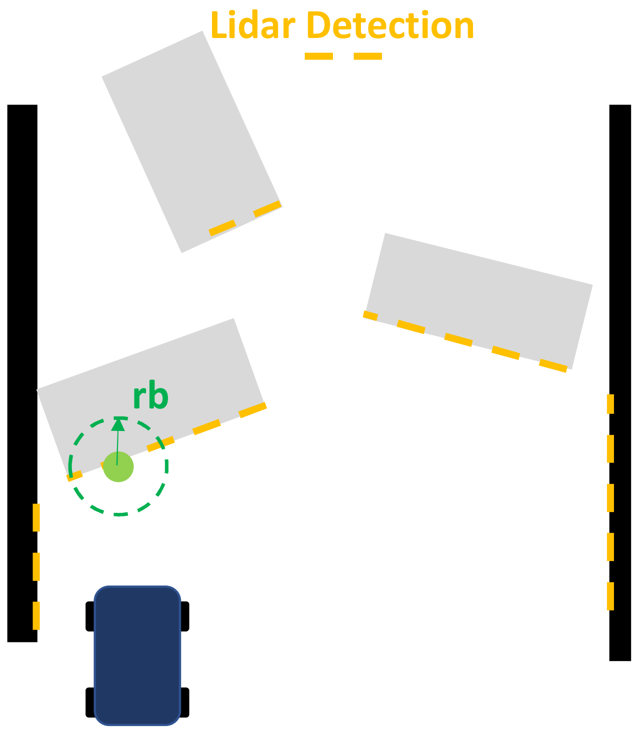


Gap: Series of at least **n** consecutive hits that pass some distance threshold **t**

n=3, t = 5.0

Follow the Gap

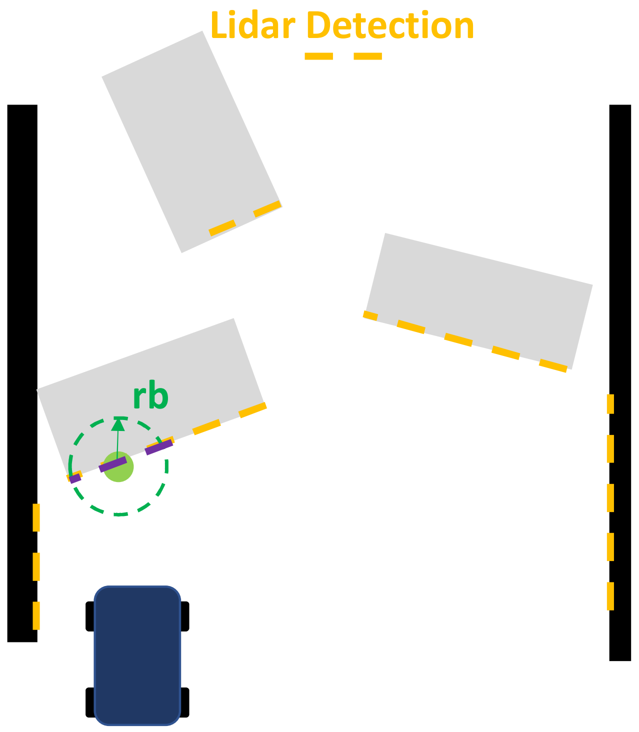
- Step 1
  - Find nearest LIDAR point and put a “safety bubble” around it of radius ‘**rb**’



[4.8, **3.1**, 4.3, 4.5, …, 10.1, 8.3]

Follow the Gap

- Step 2
  - Set all points inside bubble to distance 0. All nonzero points are considered ‘**free space**’



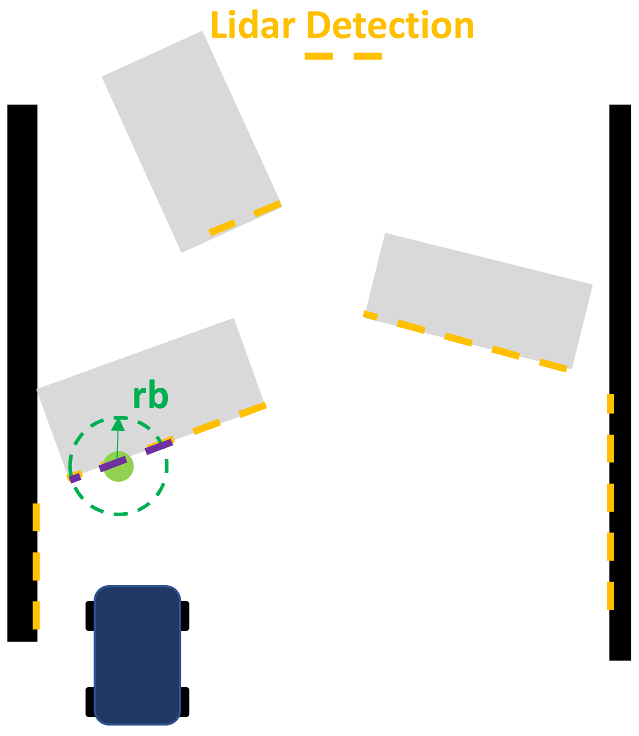
[4.8, **3.1**, 4.3, 4.5, …, 10.1, 8.3]

↓

[4.8, **0.0**, **0.0**, **0.0**, …, 10.1, 8.3]

Follow the Gap

- Step 3
  - Find maximum length sequence of consecutive non-zeros among the ‘**free space**’ points. (The **max gap**)



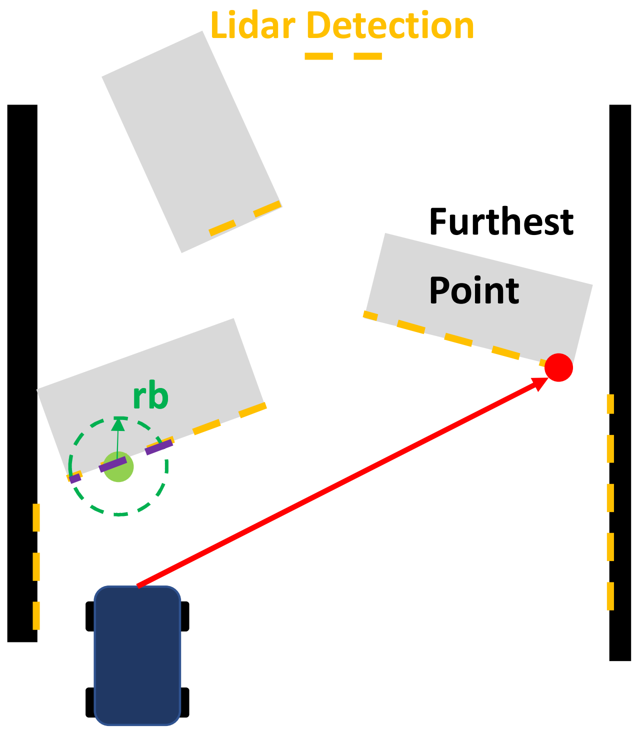
[4.8, **3.1**, 4.3, 4.5, …, 10.1, 8.3]

[4.8, **0.0**, **0.0**, **0.0**, …, 10.1, 8.3]

Follow the Gap

- Step 4
  - Find the ‘**best**’ point among this maximum length sequence

  - **Naïve**: Choose the furthest point in free space, and set your steering angle towards it



[4.8, **3.1**, 4.3, 4.5, …, 10.1, 8.3]

[4.8, **0.0**, **0.0**, **0.0**, …, 10.1, 8.3]

In summary

- Obtain laser scans and preprocess them by replacing NaN/Inf values and clipping ranges to a maximum distance.
- Find the closest point in the LIDAR ranges array and place a safety bubble of radius **rb** around it
- Set all points inside this bubble to distance 0. All remaining non-zero points are now considered "**free space**".
- Find the max length gap, in other words, the longest sequence of consecutive non-zero elements in your ranges array.
- Find the **best goal** point in this gap. Naively, this could be the furthest point in free space, and set your steering angle towards it.

Gap Following Implementation

$ cd ~/creating_autonomous_car_ws/src/creating_autonomous_car
$ code ./stack_master/config/reactive_controller.yaml

Parameters for gap following

Tune this .yaml !!

```yaml
gap_follow:
  ros__parameters:
    control_rate_hz:  50.0
    estop_min_dist:   0.5    # [m]
    estop_angle_deg: 30.0    # [deg]
    gf_bubble_radius: 0.3    # [m]
    gf_speed:         2.0    # [m/s]
    gf_max_steer:     0.4    # [rad]
    gf_max_range:    10.0    # [m]

wall_follow:
  ros__parameters:
    control_rate_hz:  50.0
    estop_min_dist:   0.5    # [m]
    estop_angle_deg: 30.0    # [deg]
    wf_target_dist:   0.8    # desired gap from right wall [m]
    wf_kp:            1.0
    wf_kd:            0.1
    wf_speed:         1.5    # [m/s]
    wf_max_steer:     0.4    # [rad]
    wf_lookahead:     0.5    # [m]
    wf_max_range:    10.0    # [m]
```

$ cd ~/creating_autonomous_car_ws/src/creating_autonomous_car
$ code ./controller/controller/gapfollow.py

Default Parameters (.yaml will override this params)

```python
PARAMS = {
    'control_rate_hz':  50.0,
    'gf_bubble_radius':  0.3,
    'gf_speed':          2.0,
    'gf_max_steer':      0.4,
    'gf_max_range':     10.0,
}
```

Gap Following Implementation

```python
def _compute(self):
    # TODO: Implement Follow-the-Gap (FTG) using LiDAR free-space selection
    #
    # Goal:
    #   - Use LiDAR scan data to find a collision-free gap
    #   - Select a safe target direction inside the best gap
    #   - Convert the target direction into a steering command
    #
    # You should return (steering, speed) from this function.
    #
    # Useful information:
    #   - self.scan.ranges             : LiDAR distance array [m]
    #   - self.scan.angle_min          : angle of first beam [rad]
    #   - self.scan.angle_increment    : angular step between beams [rad]
    #   - self.bubble_radius           : safety bubble radius around closest obstacle [m]
    #   - self.max_range               : cap LiDAR ranges to suppress outliers [m]
    #   - self.max_steer               : steering clamp [rad]
    #   - self.speed                   : nominal driving speed [m/s]
    #   - self.odom                    : current vehicle motion (optional for speed adjustment)
    #
    # Suggested approach (FTG pipeline):
    #   - Preprocess ranges:
    #       * replace NaN/Inf/invalid values
    #       * clip ranges to [0, self.max_range]
    #       * optionally focus on a front field-of-view
    #   - Find the closest obstacle beam
    #   - Create a safety bubble around that obstacle (zero-out nearby beams)
    #   - Find the longest contiguous non-zero gap
    #   - Choose the best target beam in the gap
    #       * e.g., farthest beam or weighted by distance and heading
    #   - Convert target beam index to steering angle
    #   - Clamp steering to +/- self.max_steer
    #   - Optionally reduce speed when |steering| is large or obstacle is close
    #
    # Output:
    #   - steering [rad]
    #   - speed [m/s]
    return 0.0, 0.0
```

Implement Gap Following Algorithm Here !

How to test Gap Following Algorithm

Launch Racecar Simulation → Test on SIM

```
# sim:=true -> simulation env option
ros2 launch stack_master low_level.launch.xml map:=f sim:=true
```

You can add 'use_estop' option!!

Launch low level on Car → Test on Car

```
# sim:=false -> Sensor env option
ros2 launch stack_master low_level.launch.xml
```

Launch Reactive Algorithm → Turn on GapFollowNode

```
# mode:=gap -> gap_follow mode
ros2 launch stack_master reactive.launch.xml mode:=gap
```

Tips

- Use **laserscan.msg** for **Debugging**
- Use **track width** for **fast** driving
- Simulation is useful for initial testing, but it is highly recommended to test on the real car as early as possible, since vehicle dynamics can behave differently from simulation.
- Never divide by very small values. If you do, certain outputs will oscillate rapidly and grow extremely large. Always analyze your code for any division operations, and clamp the denominator to prevent division by values below a safe minimum threshold.

(Announcement) Assignment 2 (*on April 9th*)

Week	Tue	Thu
1	Course Introduction	Intoduction to Autonomous Driving System
2 (**Team-up**)	Coordinate Frames	ROS2 Tutorial- Write a Publisher and Subscriber
3	Hardware & ROS2 Practice	Hardware & ROS2 Practice
4	Vehicle Dynamics and Control	Emergency Stop Implementation
5	Wall Following Algorithm Implementation **(Group Assignment 1)**	Wall Following Algorithm Evaluation - **(Due of Group Assignment 1)**
6	Gap Following Algorithm Implementation **(Group Assignment 2&3)**	Gap Following Algorithm Evaluation - **(Due of Group Assignment 2)**
7	Practice for Reactive Algorithm Evaluation	**Reactive Algorithm Evaluation (with obstacles)** - **(Due of Group Assignment 3)**
8	Midterm Week	
9	SLAM1	SLAM2 **(Assignment 4)**
10	Holiday (Children's Day)	SLAM & Particle Filter Practice
11	Path Planning Technique 1 - RRT	Path Planning Technique 2 - Global Optimization **(Due of Assignment 4)**
12	Path Following Technique – Pure-Pursuit **(Assignment 5)**	Lidar Object Detection and Tracking **(Assignment 6)**
13	Planning - Control Practice **(Due of Assignment 5)**	Path Planning Technique - Spline based Local Planning **(Due of Assignment 6)**
14	Final Racing Practice (ICRA 2026)	Final Racing Practice (ICRA 2026)
15	Final Racing Practice	Final Racing Practice
16	Final Racing Practice	**Final Racing Grand Prix**

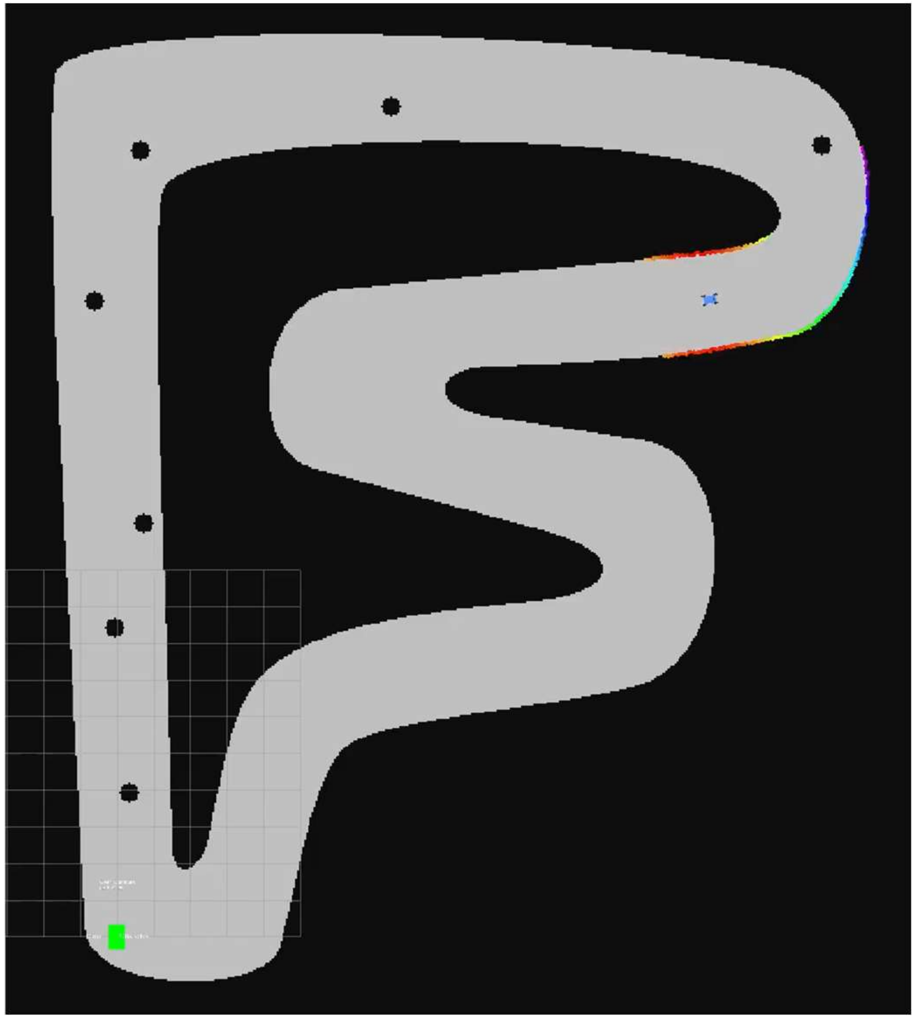
Assignment 2 :

Test track driving using a gap-following algorithm (On Car) → **Pass / Fail**

- Complete the 3-laps without any collision

Assignment 2

- The goal of Assignment is to develop a reactive algorithm for race car to avoid obstacles.
- Find and follow the drivable gap.
- Imagine how the vehicle's behavior changes as speed increases or decreases, then consider how each parameter or strategy should be adjusted so that gap following can still avoid collisions even at high speeds.



(Announcement) Assignment 3 (*on April 16th*)

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Assignment 3 :

Evaluate track driving at ***high speed*** using **reactive algorithms** on the track → **Total 10 points**

- **Reactive Algorithm Evaluation** - Test **reactive algorithm** with **high speed** (**No obstacles** on track)
- **Reactive Algorithm (With Obstacles)** - Test **reactive algorithm** with **high speed** (**Obstacles** on track)

UNIST

HMCL High-assurance Mobility Control Lab

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16	Final Racing Practice	**Final Racing Grand Prix**

Assignment 3 :

Evaluate track driving at ***high speed*** using **reactive algorithms** on the track → **Total 10 points**

- Complete the 3-laps without any collision → 5 points per test / 1 point penalty per collision for each test
- Fastest lap → Extra 1 points per test

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Assignment 3 :

Evaluate track driving at ***high speed*** using **reactive algorithms** on the track → **Total 10 points**

- Each group has **2 trials** per test
- Completing **3 laps** guarantees at least **2 points**

Assignment 3

2025 PNU F1TENTH Competition